• Measurement of physical properties, such as temperature, velocity, displacement, strain and more
• Real-time monitoring of health
• Testing of buildings, bridges, tunnels, dams, heritage structures and so on
• Night-vision cameras, electronic security systems and in measuring wheel loads of vehicles

Industrial elements—such as furnaces of all kinds, sintering operations, ovens and kilns, automated welding and more—often generate large electrical fields where conventional sensors cannot be used. High temperature processing operations, as in refractories and chemical industries, often use optical-fibre sensors. These are also used in fusion, sputtering and crystal growth processes in semiconductor industries.

***Oil and gas industry.*** The oil and gas industry often faces huge problems, and needs effective and safe solutions from time to time. For measuring temperature and pressure developed in the downhole of oil wells, ordinary sensors cannot be used. So, fibre-optic sensors can come to the rescue here.

Hydraulic fracturing is needed for the highest potential success of oil and gas wells. This can be effectively done using fibre-optic sensors.

Accurate downhole monitoring, data collection and analysis with minimal environmental impact are needed to increase the life of oil and gas wells. This can be easily achieved using fibre-optic sensors.

For permanent downhole installations, optical-fibre cables are protected inside hermetically-welded metal tubes that are typically filled with special buffer gels. Here, fibre-optic sensors can provide useful data in less time, which can bring higher automation to the oil and gas industry.

***Marine industry.*** Epsilon Optics provides all fibre-optic strain sensing for Oracle racing. Virtually all critical structures are monitored, including hulls, foils and rigs, providing vital feedback to designers and reassurance to the sailing team. Marine applications for fibre-optic sensors include life boats, fast surface crafts, luxury yachts, high-performance racing yachts, naval vessels and submarines.

## Using fibre-optic sensors as a universal medium of service

Despite the deep penetration of smart sensors and wireless sensors, fibre-optic sensors are still popular. After extensive field research, industrialists are now realising that fibre-optic sensors are the only alternative for hazardous industrial situations.

These sensors use light for measuring a quantity. Quantity is measured by modulation on intensity, spectrum, phase or polarisation of light travelling through optical fibre. It also makes use of optical sensors that convert light waves into electronic signals, which can be easily read by instruments.

Advantages of fibre-optic sensors for industrial process control instrumentation are significant. Cost of these sensors is continuously falling, while technical performance is improving. Fibre-optic sensors are ideal for use in refineries, chemical plants, power plants, oil production refineries or in any such hostile environment, since these are safe to be used in hazardous situations.

## Future trends to the value chain

These days, fibre-optic sensors come with fibre-optic amplifiers, with easy-to-read digital LEDs. These digital displays provide real-time feedback. Newer sensors need significantly less wiring. For instance, in a configuration, there must be 16 sensors, all connected and sharing a single power line.

Newer sensors incorporate either a 12-bit or 16-bit CPU along with a 12-bit A/D converter that provides both higher resolution and faster response time. Certain models of sensors allow users remote access. This is especially useful in potentially hazardous environments. Using remote controllers, sensors in hostile environment can be programmed and monitored from a safe distance.

With all the flexibility and benefits, fibre-optic sensing systems can be widely used. Particularly, these find potential applications in industrial automation.

Fibre-optic sensors are mainly used in remote-sensing applications. These are resistant to electromagnetic interferences and do not conduct electricity. Hence, these find applications that involve highly-inflammable materials or high-voltage electricity.

Future trends in the fiber-optic sensors field can be briefly summarised as follows:

• Special wave guides, such as photonic crystal fibres, will enable many new sensing mechanisms and sensor configurations.
• Improved micro-fabrication technologies will continue to improve sensor performance, functionality and reliability.
• Advanced signal processing and networking technologies will enable high-density fibre-optic sensor networks.

As a part of technological initiatives, nanotechnology is slowly gaining momentum. Recent research has presented a cost-effective and rugged approach to apply nanoparticles to optical-fibre cables. Use of nanoparticles on coatings of fibre-optic cables leads to enhanced sensor response. These particles provide several advanced abilities to sensors, such as real-time monitoring, multiplexing capabilities, remote sensing, small dimensions, bio-compatibility and much more.

The main goal of these newer trends is to gain more benefits from these sensors.

Plasmotronics is yet another promising new trend in electronics. Researchers have found that plasmons can be effectively used in optical-fibre cables to enhance their capabilities. Here, plasmons are used instead of photons (light particles). Thus, fibre-optic plasmonic sensors utilise both propagation and localised surface plasmon resonance techniques.

## All-in-all

Fibre-optic sensors can be effectively integrated with instrumentation functions. In this era of communication and connectivity, these find many applications. These sensors are emerging as a practical solution for many industrial applications, being faster, more efficient and effective for electronic instrumentation. EFY